20XT96 - MATHEMATICAL MODELLING LAB

GENOMEFOLIO — PACKAGE ABSTRACT

Team members:

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Introduction

Financial return series are usually grouped with correlation. Correlation measures how much two series move together over a whole window, but it cannot recognise the same *pattern* when it appears a few days earlier in one stock and a few days later in another. This project instead treats each stock's return history as a **symbolic sequence**, groups stocks by **sequence similarity**, and forecasts each group's behaviour.

Description

GenomeFolio implements a five-stage pipeline.

1. **Data** — daily adjusted prices for a fixed universe of US large-cap stocks are converted to simple daily returns on their common trading days.
2. **Symbolization** — each stock's returns over the trailing training window become a five-symbol alphabet: big-down, small-down, flat, small-up, big-up. Thresholds are per-stock quantiles of the training window, so one stock's volatility does not dominate the comparison.
3. **Alignment** — every pair of symbol strings is aligned with a banded **Needleman-Wunsch** global alignment using affine gaps and a graded substitution matrix. The band limits how far a pattern may shift, so unrelated days cannot be paired.
4. **Clustering** — the normalised alignment scores form a distance matrix, which is clustered with hierarchical average linkage. The number of clusters is chosen by silhouette under a constrained search with a minimum cluster size.
5. **Forecast** — each cluster's equal-weight composite return is forecast over the coming 21 trading days with ARIMA; a spline model is available as an alternative.

The pipeline stops at the forecast. It deliberately contains no null-hypothesis gate, no portfolio construction and no backtest.

Datasets Used

- **Primary** — 20 fixed cross-sector US large-caps, daily adjusted close from 2013 onward, downloaded once and cached locally.
- **Offline fixture** — a committed 10-stock sample covering five years, used by the tests and by the offline demo so the project runs without network access.

Method and Implementation

The implementation is pure Python 3.11 with NumPy, pandas, SciPy, scikit-learn and statsmodels; the banded alignment dynamic program is compiled with numba. Every parameter lives in `config/params.yaml`; nothing is hardcoded. Each stage uses only data up to the point being processed, so there is no look-ahead.

The distance between two symbol strings is the alignment score normalised by the two self-scores, `1 - score / sqrt(self_a * self_b)`. It is treated as a similarity measure, not as a mathematical metric: the code asserts symmetry and non-negativity, which is all average linkage requires.

Expected Outcome

The expected outcome is (a) interpretable groups of stocks whose return patterns are structurally similar and (b) a 21-day forecast for each group's composite. Because daily returns are only weakly predictable, the forecast is reported as a modest signal with its horizon, and limitations — survivorship, window sensitivity and weak predictability — are documented rather than hidden.

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